

Digital Organizational Resilience as a Socio-technical Capability

Abstract

Resilience has become a central concern in management and information systems as organizations confront increasingly complex digital disruptions. Despite its prominence in policy, standards, and organizational discourse, resilience often has remained highly rhetorical and little operational: Conceptual ambiguity, weak operationalization, and fragmented empirical evidence hinder the development of proper organizational foundations for resilience, which in turn hinders the deployment of actionable practices. This paper addresses these challenges through a structured, cross-disciplinary review of resilience research across management, information systems, and related fields. We identified three core limitations in the current understanding of digital resilience in organizations: weak construct clarity, insufficient indicators capable of capturing adaptive performance when disruptions occur, and limited accumulation and exploitation of evidence that would help measure resilience and act upon across organizational contexts. Here, we recognize that digital organizational resilience shall be an enacted socio-technical capability linking organizational routines, technological infrastructures, and ecosystem interactions. We further outline a research agenda that emphasizes early-warning systems, integrative design principles, measurable behavioral indicators, and the importance of combining field data and simulations. This study contributes to theory development and provides a foundation for de-

veloping observable and actionable forms of resilience in complex digital environments.

Keywords: Digital Organizational Resilience, socio-technical Systems, Organizational Adaptation, Digital Infrastructure, Resilience Measurement.

1 INTRODUCTION

In early 2023, engineers at Samsung inadvertently exposed proprietary source code by entering confidential information into publicly accessible generative artificial intelligence (AI) tools, prompting the company to ban the use of ChatGPT and similar systems among employees to prevent further leaks (Ray, 2023). This incident was not an isolated anomaly. Since then, large organizations have reported a growing number of generative-AI-related data exposures and security violations (Arora, 2025). More broadly, advanced AI systems are expected to automate and accelerate cyberattacks, lowering the entry barrier for sophisticated threats and enabling malicious actors to scale operations in ways previously constrained by human and technical resources (Arif et al., 2024).

These developments reflect a broader shift in the conditions under which organizations operate. As digital technologies have become more deeply embedded in organizational processes, coordination, and interdependence, disruptions increasingly arise not only from technical failures, but also from human practices and their interaction with technical systems (Dupont, 2019). Such disruptions cannot be addressed through technical safeguards or formal policies alone. They require organizations to build capabilities for adaptation and coordination across intertwined human and technical components under conditions of uncertainty (Boin & van Eeten, 2013). Resilience therefore concerns the organizational ability to adjust, reorganize, and sustain operations as disturbances unfold (Burnard & Bhamra, 2011).

Against this backdrop, resilience has become a central regulatory priority (Heinimann & Hatfield, 2017). Crises such as COVID-19 (Almeida et al., 2020), large-scale cyber

incidents (Dupont, 2019), and systemic information technology failures (Pumpuni-Lenss et al., 2017) have transformed resilience from an abstract organizational aspiration into a governance mandate (Boin & van Eeten, 2013). Across sectors ranging from finance to healthcare and critical infrastructure, organizations are increasingly expected to anticipate, absorb, and adapt to disruption while maintaining operational continuity and legitimacy. This expectation is reinforced by landmark regulatory initiatives such as the European Union's Digital Operational Resilience Act (DORA) and Australia's Security of Critical Infrastructure Act 2018, which formalize resilience requirements in sectors exposed to digital and cyber disruption (Commonwealth of Australia, 2018; Parliament & of the European Union., 2022). Yet practitioners have repeatedly expressed concern that most organizations are not designed to adapt quickly enough to an accelerating and unstable technological environment (Pariès & Wreathall, 2017). A severe mismatch therefore persists between regulatory ambition and organizational capability: managers still struggle to define what resilience means for their organization, let alone how to measure and operationalize it (Björck et al., 2015). More fundamentally, organizations increasingly face disruptions that are continuous, novel, and overlapping, whereas many organizational structures and assurance systems remain oriented toward a limited set of exceptional shocks. One consequence is that audit-driven compliance and rigid control frameworks often privilege documentation and formal assurance over the development of adaptive capacity (Power, 1999; Woods, 2015). They may signal preparedness, yet often do little to strengthen learning, coordination, and flexible response under persistent uncertainty.

These tensions are not confined to the literature; they are also visible in organizational practice. To complement the review, we draw on an exploratory qualitative study based

on 18 semi-structured interviews with cybersecurity and risk professionals from large organizations in Switzerland, selected for their involvement in cyber risk management and organizational resilience issues (see Appendix C for the full study design, sample characteristics, interview protocol, and detailed results). This material revealed six recurring observations (see Table 1): (i) persistent ambiguity in defining and quantifying resilience, (ii) the dominance of regulatory pressure and symbolic compliance (Power, 1999), (iii) the persistence of siloed organizational structures, (iv) the limited use of incidents as opportunities for organizational learning, (v) the uneven integration of human and cultural factors, and (vi) the tension between regulatory demands for flexibility and organizational demands for precision. These observations reinforce a central point of this review: resilience is widely prescribed, but only unevenly enacted as an organizational capability.

Theme from Field Evidence	Dimension of Resilience in Organization	Implications for Practice
Quantification & Ambiguity	Organizations lack operational metrics; resilience remains rhetorical and fragmented across units.	Shared definitions and standardized constructs are needed to enable measurement.
Regulatory Pressure & Symbolic Compliance	Regulatory audits dominate, producing “tick-box” resilience without adaptive capacity.	Tools must assess adaptive behavior, not just documentation.
Coordination & Silos	IT, risk, compliance units operate in silos; limited cross-functional coordination.	Organizational design should support cross-unit collaboration under crisis.
Learning from Incidents	Lessons learned rarely institutionalized; ad-hoc reviews lack persistence.	Mechanisms are needed to embed feedback and retain knowledge.
Human & Cultural Factors	Expertise, simulation exercises, and culture matter, but interpretations vary across units.	Behavioral and cultural aspects should be integrated into resilience frameworks.
Flexibility–Precision Paradox	Regulators provide flexible definitions; firms request precise ones.	Balance between contextual adaptation and measurable guidance must be achieved.

Table 1: Field evidence on how organizations interpret and enact resilience in practice (Method : semi-structured interviews, c.f. Appendix C for further results and methodological aspects).

The challenge of resilience is compounded by conceptual ambiguity in the scientific literature. As resilience has developed across disciplines and industries, its meaning has fragmented, ranging from system uptime to business continuity, regulatory posture, and security architecture, thereby undermining conceptual coherence and practical accountability (Hillmann & Guenther, 2021; Woods, 2015). Even where organizations adopt established resilience and risk-management practices – such as audits, business continuity planning, incident response procedures, or post-incident reviews – incidents and near-misses are seldom leveraged as sustained opportunities for organizational learning, leaving reflexivity, decentralization, and coordination underdeveloped (Barasa et al., 2018;

Francis & Bekera, 2014). A growing number of standards and guidelines now address resilience (Linkov & Kott, 2018), yet organizations still lack comprehensive methods to quantify, simulate, design, and implement it, creating the risk that resilience loses traction in practice despite its growing importance (Kuypers & Maillart, 2018). These difficulties are compounded by the fact that resilience has been translated into organizational settings through partially divergent assumptions about stability, adaptation, control, and transformation.

Advancing resilience in organizational settings therefore requires a clearer understanding of resilience as a socio-technical capability. To address this challenge in management research and practice, we adopt an integrative and problem-driven approach. Rather than adding another abstract definition or prescribing generic best practices, we examine why resilience repeatedly collapses into compliance, symbolic adoption, or fragmentation when organizations attempt to translate it into practice. By tracing how resilience has been conceptualized, operationalized, and enacted across disciplines – and where these translations fail – we shift the analytical focus from what resilience ought to mean to how it can be made observable and actionable in organizational settings.

The remainder of the paper proceeds as follows. Section 2 reviews how resilience has been conceptualized across disciplines and translated into management and information systems research. Section 3 identifies the conceptual, methodological, and empirical gaps that help explain why resilience remains difficult to enact in practice. Section 4 introduces the enactment-scope framework for digital organizational resilience. Section 5 develops a research agenda centered on measurable indicators, simulation-based approaches, and

integrative frameworks. Section 6 discusses the implications for IS research and organizational practice. Finally, Section 7 concludes by outlining pathways toward cumulative and actionable resilience research in the IS field.

2 BACKGROUND

This section examines how resilience has been conceptualized across disciplines and how those ideas were later taken up in management and information systems research. Resilience did not travel from one field to another as a stable concept. It developed through distinct intellectual traditions, each carrying its own assumptions about stability, adaptation, control, and system change. Those differences matter in organizational settings because they shape what resilience is taken to mean, how it is measured, and what kinds of interventions are treated as legitimate or effective. Tracing these differences helps explain why resilience has become so widely invoked in organizations while remaining difficult to define, compare, and operationalize consistently.

2.1 How Resilience Was Shaped Across Disciplines

Mathematics and Physics The earliest foundations of resilience emerged in mathematics and physics, where it was generally understood as the capacity of a system to absorb disturbance and return to equilibrium. Early formulations in physics and systems thinking emphasized elasticity, stability, and homeostatic feedback as defining features of resilient behavior (Panteli et al., 2017; Rankine, 1858; Wiener, 1948). In that view, resilience is mainly about rebound and temporal recovery. This perspective proved especially useful for formal and quantitative treatments of engineered systems because disturbance-response dynamics can be specified with considerable precision. Its limitation is equally clear: it

leaves relatively little room for cases in which systems do not return to a prior stable state.

Biology and Ecology Biology and ecology changed the meaning of resilience more fundamentally. The emphasis moved away from equilibrium restoration and toward persistence through variability, adaptation, and regime change. Resilience no longer meant return to a prior state so much as the capacity to reorganize while preserving core functions (Gunderson & Holling, 2002; Holling, 1973). This perspective stresses that resilience emerges through feedbacks across ecological, social, and institutional levels (Sundstrom et al., 2014). Ecological models such as the adaptive cycle illustrate transitions between stability regimes and show how diversity and redundancy can buffer systems against collapse (Folke et al., 2004, 2010; Walker et al., 2004). Early metrics such as return variance, autocorrelation, and recovery time were used to characterize resilience under uncertainty (Ives, 1995) (see Appendix A.2). This tradition matters because it treats resilience as persistence through reorganization rather than simple recovery. At the same time, it creates a problem for later organizational research: once resilience includes transformation as well as restoration, the relevant baseline, success criteria, and performance measures become harder to specify.

These perspectives are often treated as compatible, but they are not built on the same assumptions. Engineering resilience is oriented toward recovery to a known and valued state, whereas ecological resilience accepts irreversibility, nonlinear change, and the possibility that persistence may require transformation rather than restoration (Holling, 1996; Walker et al., 2004). That tension becomes difficult to ignore in organizational research, where resilience is often invoked as if stability, adaptation, and transformation formed a single coherent objective (Duchek, 2019; Woods, 2015).

Neuroscience and Psychology Research in neuroscience and psychology extends the concept again by framing resilience as the capacity of individuals and groups to endure, adapt, and sometimes grow through adversity (Kalisch et al., 2024; Luthar et al., 2000). This work emphasizes mechanisms such as cognitive flexibility, learning, and emotional regulation as central to adaptive responses under stress (Feder et al., 2019; Masten & Osofsky, 2010). Empirical studies further connect resilience to stress-regulation processes and to the plasticity of adaptive capacities shaped by experience over time (McEwen, 2004; Rakesh et al., 2019). Psychometric instruments such as the Wagnild and Young Scale (Wagnild & Young, 1993) and the Connor-Davidson Resilience Scale (Campbell-Sills & Stein, 2007) operationalize resilience mainly at the individual and group levels. Moving beyond strictly individual accounts, Ungar (2021) treats resilience as a multisystemic phenomenon distributed across personal, social, institutional, and ecological contexts. In organizational settings, especially in domains such as cybersecurity operations, healthcare, and emergency response, these insights help explain how stress, judgment, coordination, and learning shape adaptive action under pressure (Haldane et al., 2021; Kruk et al., 2017; Norris et al., 2007). What remains less developed is the link to socio-technical organization research. Individual resilience has been studied extensively, but far less is known about how these adaptive capacities scale to collective organizational processes or interact with technological infrastructures.

Institutional Economics A major exception to individual-centered accounts appears in institutional economics, which places the collective and governance dimensions of resilience at the center. In her foundational analysis of common-pool resources, Ostrom (1990) showed how communities operating under uncertain and demanding conditions can sustain shared resources through self-organized governance arrangements grounded

in trust, reciprocity, monitoring, and graduated sanctions. In this tradition, resilience is not merely recovery or endurance. It is the capacity of a socio-economic system to sustain cooperation under uncertainty, external pressure, and resource constraints. Research on social-ecological systems similarly shows that long-term resilience depends on adaptive rule-making, polycentric governance, and alignment between incentives, actors, and shared resource systems (Lebel et al., 2006; Ostrom, 1990). This tradition is important because it treats resilience as an emergent outcome of institutional design and collective action rather than as an individual trait or a purely technical property. What it leaves more open is how these governance principles should be translated into contemporary organizations whose coordination increasingly depends on digital infrastructures and interorganizational dependencies.

Cybernetics and Engineering Control theory and engineering reintroduced resilience into applied domains such as infrastructure, safety, and risk management. Here resilience was typically defined as the ability of a system to recover functionality after disruption, often measured through recovery time, reliability, and functional uptime (Bruneau et al., 2003). A foundational engineering contribution distinguishes resilience as recovery speed, reliability as frequency of success, and vulnerability as severity of failure (Hashimoto et al., 1982). This equilibrium-oriented view remains influential where rapid restoration of critical functions is the primary objective (see Appendix A.1). In resilience engineering, Pariès and Wreathall (2017) identified four practical capabilities – responding, monitoring, learning, and anticipating – as key levers for improving system resilience. This tradition offers strong operational clarity and a clear evaluative logic. Its limits become more visible, however, in settings marked by adversarial adaptation, non-stationary environments, and strategic contestation, such as cybersecurity and digitally networked systems (Dupont,

2019).

Complexity and Socio-technical Systems Complexity science and socio-technical systems theory shift the focus once more by emphasizing emergence, self-organization, and nonlinear interaction. In this view, resilience arises less from centralized control than from local interaction among heterogeneous actors and from feedback mechanisms operating across scales (Krakovská et al., 2023; Levin, 1998). Modeling traditions such as agent-based simulation, network analysis, and game-theoretic approaches help explain how system-level resilience can emerge from distributed behavior (Albert et al., 2000; Gao et al., 2016; Myerson, 2013). Research on network robustness, for example, shows that scale-free systems may be resilient to random failures yet highly vulnerable to targeted attacks (Albert et al., 2000). Cross-scale work on infrastructure further shows that resilience must be understood across spatial, temporal, and governance levels because disruptions frequently cascade across them (Moreno & Shaw, 2018). Social-ecological systems research complements this perspective by stressing the role of institutions, collective learning, adaptive governance, and actor-resource-rule configurations in sustaining resilience (Adger et al., 2005; Anderies et al., 2004; Berkes et al., 1998). The structure of social networks matters as well because it conditions information flows, coordination capacity, and collective action (Bodin & Crona, 2009). More recent work also makes clear that resilience strategies entail political, social, and economic trade-offs (Aase et al., 2020; Leichenko & O'Brien, 2024; Sornette et al., 2013). This perspective is especially useful for organizational and digital contexts because it foregrounds distributed adaptation and interdependence. Its drawback is different: it is often easier to identify resilience as an emergent system-level property than to translate that property into clear organizational responsibilities, interventions, or metrics.

These traditions do not converge on a single understanding of resilience. Some define it as recovery to equilibrium, others as persistence through reorganization, and others as a function of cognition, governance, or distributed interaction. Resilience therefore entered organizational research not as a unified construct but as a composite label carrying partly incompatible assumptions about what should persist, what may change, who adapts, and how adaptation should be evaluated. Those tensions are central to understanding why resilience became attractive in organizational discourse while remaining difficult to define and operationalize consistently.

2.2 Resilience in Organizations

Organizational research brought resilience much closer to managerial practice. In this literature, resilience is no longer treated only as a broad system property. It is translated into questions of capability, design, governance, and adaptation. That translation was productive, but also selective. Management and organization scholars drew on several resilience traditions at once without fully resolving their underlying differences, especially over what resilience consists of, how it should be measured, and whether it refers mainly to stability, adaptation, or transformation.

2.2.1 The Construction of Resilience in Organizations

A large part of the organizational literature explains resilience through two broad sets of mechanisms: structural arrangements and cultural practices. Together, these strands describe how organizations prepare for disruption, respond under pressure, and learn over time (Bhamra et al., 2011; Ducheck, 2019; Weick & Sutcliffe, 2011).

Structural arrangements matter because they shape how control, information, and resources move when conditions become unstable. Decentralized structures can support quicker local responses when disruptions unfold unevenly across units or time horizons (Boin & van Eeten, 2013). Modularity can limit cascading failure by isolating breakdowns and preserving partial system functionality. Ambidextrous forms of organization can also help firms maintain efficiency while preserving enough flexibility to innovate under pressure (Bhamra et al., 2011). These features are especially relevant in digital and cybersecurity settings, where overly centralized coordination can create bottlenecks and increase vulnerability.

Cultural practices matter no less. Rehearsals, simulation-based training, and pre-mortem analysis help build shared mental models, reduce uncertainty, and strengthen trust under stress (Klein, 2008; Weick & Sutcliffe, 2011; Westrum, 2004). Practices such as improvisation, learning orientation, and psychological safety further support teams facing novel or high-pressure situations (Lanzara, 1999; Vogus & Sutcliffe, 2007). In fast-moving threat environments, these practices make it possible to react and learn before formal control systems can be revised (Duchek, 2019; Westrum, 2004).

One of the main contributions of this literature is that it makes resilience organizationally interpretable. Resilience is no longer treated simply as a system-level outcome, but as something that can be shaped through design choices, routines, and cultural conditions. The difficulty is that many of these structural and cultural enablers are linked to resilience without clearly identifying the pathways through which they produce adaptive behavior or the indicators that would show whether such behavior is actually taking place.

2.2.2 Resilience as a Dynamic Managerial Capability

A second major line of work treats resilience as a dynamic managerial capability. From this perspective, resilience is less a reactive outcome than a path-dependent capacity formed through routines, governance arrangements, resource deployment, and strategic intent (Boin & van Eeten, 2013; Lengnick-Hall et al., 2011). This marks a shift away from narrow views of robustness or risk avoidance. The emphasis falls instead on an organization's ability to interpret uncertainty, reconfigure resources, and continue functioning under pressure.

Recent work explicitly frames organizational resilience as a form of dynamic capability that allows firms to sense disruption, respond effectively, and in some cases derive advantage from it (Duchek, 2019; Teece, 2007). In that view, resilience depends on microfoundations such as routines, heuristics, and cognitive schemas that shape how actors interpret events and act under uncertainty (Hepfer & Lawrence, 2022). Resilience therefore cannot be reduced to resistance or continuity alone. It requires an adaptive orientation embedded in both operational and strategic action.

In practice, that capability is usually supported by combinations of managerial levers. Slack resources and redundancy can provide buffering capacity (Barasa et al., 2018). Ambidextrous arrangements can support the simultaneous pursuit of stability and innovation (Bhamra et al., 2011). Resilience audits and readiness diagnostics aim to assess whether organizations can absorb and adapt to shocks (Burnard & Bhamra, 2011; Calder, 2021). This stream is useful because it translates resilience into a language managers can act on. What it still does not resolve is a basic problem: resilience is often described as a desirable

capability before there is clear agreement on what observable performance would actually demonstrate that the capability exists.

2.2.3 Cybersecurity Resilience as a Strategic and Institutional Imperative

Cybersecurity is a particularly revealing setting because many of the broader organizational tensions around resilience become more visible there. Cyber risks are marked by uncertainty, rapid change, and interdependent vulnerabilities that cut across technical, human, and organizational layers (Linkov et al., 2013). They are also often asymmetric: threats may emerge quickly, evolve during response, and remain only partly visible until damage is already underway. Under such conditions, compliance-oriented and control-oriented approaches are not sufficient on their own (Dupont, 2019). This is one reason cybersecurity is now increasingly treated not only as a control problem, but as a resilience problem (Dupont, 2019; Linkov et al., 2013).

The cyber-resilience literature generally presents resilience as an integrated capability combining technological safeguards, strategic foresight, organizational adaptability, and institutional alignment (Björck et al., 2015; Lostri et al., 2022). Standards such as ISO 22301 emphasize the maintenance of core functions under disruption (Calder, 2021), while other work places more weight on learning, reflexive governance, and ongoing adjustment (Hausken, 2020). Regulatory pressures, including those associated with Basel III and related resilience initiatives in finance, have further reinforced resilience as a strategic imperative by tying it to legitimacy, continuity, and systemic risk management (Almeida et al., 2020; Basel Committee on Banking Supervision, 2011). This has contributed to a broader shift in governance: firms are increasingly judged not only by the strength of their

own security posture, but also by the extent to which they help sustain resilience across wider ecosystems (Björck et al., 2015; Gillard et al., 2023; Lostri et al., 2022).

At the same time, this work also exposes the limits of current organizational thinking. Formal controls, standards, and regulatory mandates may clarify accountability and guide resource allocation, but they can just as easily encourage symbolic compliance and document-centered assurance rather than adaptive capacity in practice. AI, predictive analytics, and scenario modeling are often presented as tools that could improve anticipation and support more adaptive responses (Gillard et al., 2023; Kuypers & Maillart, 2018). Yet their integration into routine organizational governance remains uneven. The result is not a stable operational model of cyber resilience, but a continuing tension between resilience as regulatory requirement, resilience as managerial capability, and resilience as an emergent property of socio-technical coordination under pressure (Boin & van Eeten, 2013; Duchek, 2019; Linkov et al., 2013).

Together, management and cybersecurity research have made resilience more visible in organizational practice by identifying structural, cultural, strategic, and governance-related conditions that support adaptive action. What they still have not resolved are three underlying problems that run through the broader resilience literature: continued ambiguity in the concept itself, limited agreement on what should count as observable resilient performance, and weak connections between conceptual claims and operational measures. These issues help explain why resilience remains attractive in organizational discourse while still proving difficult to define and enact consistently in digitally mediated settings.

3 RESEARCH GAPS

Resilience is now a routine demand in policy documents, strategy discussions, and academic work, yet it remains difficult to use analytically in organizational settings. The problem is not that the literature is thin. It is the opposite. Engineering, ecology, psychology, and complex systems have all produced substantial work on resilience, but they are not describing the same phenomenon and they do not proceed from the same assumptions. Most of these traditions were not developed for organization-level analysis in the first place. Once resilience is imported into managerial and regulatory settings, that divergence stops being an abstract theoretical issue and becomes a practical one. Organizations are asked to demonstrate resilience, yet they are given limited clarity about what that should mean in practice, how it should be measured, or how it should be governed (Hillmann & Guenther, 2021).

Part of the problem is semantic, but only part. Across disciplines, resilience has been used to refer to recovery, robustness, adaptation, learning, and transformation. These labels are often treated as adjacent, sometimes even as interchangeable, yet they rest on different views of time, change, and system behavior. A recovery-oriented account assumes that disruption is followed by return. A transformative account does not. It allows for the possibility that disruption changes the system more fundamentally. When these perspectives are carried into organizational settings, they do not produce the same priorities. A firm may invest in robustness and then be evaluated on adaptability, or build compliance structures while being told it should become more transformative.

This review argues that three related problems follow from that situation: concep-

tual plurality without operational convergence, limited empirical validation of adaptive performance, and methodological fragmentation across domains. The first problem is conceptual. Management and information systems research still lacks a shared operational core for studying resilience. The second is empirical. Resilience is rarely examined through direct observation of how organizations adapt over time. The third is methodological. Existing metrics and formal models remain only weakly connected to the governance and evaluation systems organizations actually use. Of these, the issue of operationalizable and transferable measures is especially important (Woods, 2015). Similar concerns have been raised in fields such as infrastructure and healthcare (Heinimann & Hatfield, 2017; Kruk et al., 2017). Details of the review process are reported in Appendix B, and Appendix C summarizes how these tensions appear in organizational practice.

First, **conceptual plurality without operational convergence**. Resilience has been conceptualized through lenses of mechanical stability, ecological transformation, cognitive adaptability, and socio-technical flexibility (see Appendix A). Each of these perspectives offers useful and often rigorously developed insights, but their assumptions are not equivalent. Engineering resilience emphasizes restoration of predefined functions. Ecological resilience foregrounds regime shifts and adaptive reorganization. Psychological resilience centers on coping, endurance, and growth under stress. Socio-technical perspectives highlight emergence, interdependence, and distributed coordination. These perspectives can be read as complementary at a high level, but their implications for organizational design, measurement, and accountability differ substantially.

In organizational practice, this plurality often turns into definitional drift. Terms such

as resistance, robustness, recovery, agility, and adaptation are frequently used as though they referred to the same thing, even when they point to different mechanisms and temporal dynamics. As a result, resilience is often invoked symbolically through strategy documents, audits, or compliance templates rather than embedded as a coherent adaptive capability (Duchek, 2019). Without a clearer account of how these perspectives translate into observable organizational mechanisms, resilience risks remaining conceptually rich while operationally diffuse.

Second, **limited empirical validation and longitudinal grounding**. Empirical studies in cybersecurity and risk management suggest that resilience is usually assessed through compliance artifacts such as audits, maturity models, and certifications rather than through systematic observation of adaptive performance during disruption. More advanced tools, including predictive simulations and anomaly detection, are sometimes piloted in innovation units, but they remain weakly integrated into regulatory evaluation structures (Skopik et al., 2016). At the same time, informal practices such as parallel communication channels, crisis improvisation, and decentralized escalation often prove crucial to adaptive response, yet remain largely invisible within existing assessment frameworks.

The same mismatch appears in healthcare and public health systems. Resilience is often practiced there, but only partially built into measurement frameworks (Kruk et al., 2017). Longitudinal studies tracing how resilience evolves across repeated disruptions remain relatively rare. Few frameworks directly address the uncertainty, reversibility, and trade-offs involved in resilience assessment (Allen et al., 2018). Disciplinary fragmentation compounds the problem. Analytical methods developed in control theory or complex

systems rarely migrate into organizational research, while widely used practices such as awareness campaigns often fail to produce measurable behavioral adaptation (Bada et al., 2019).

To examine how resilience research is distributed across domains and methodological approaches, we systematically classified 89 publications according to thematic focus and analytical method. The resulting cross-disciplinary map (Table 2) shows clear asymmetries in how resilience is studied, particularly the limited use of quantitative markers, metrics, and simulation approaches within organizational and managerial contexts.

Topics		Methodologies						
		Simulation & Modeling	Network Analysis	Markers & Metrics	Technology	Case Studies	Longitudinal Studies	Surveys
Complex Adaptive Systems		Holling, 1973 Gao et al., 2016 Pumpuni-Lenss et al., 2017 Krakovská et al., 2023	Albert et al., 2000 Bodin and Crona, 2009 Gao et al., 2016 Pumpuni-Lenss et al., 2017	Anderies et al., 2004 Sundstrom et al., 2014 Pumpuni-Lenss et al., 2017		Ostrom, 1990 Lanzara, 1999 Folke et al., 2004 Weick and Sutcliffe, 2011		Ostrom, 1990 Levin, 1998 Gunderson and Holling, 2002 Walker et al., 2004 Heinimann and Hatfield, 2017
Engineering		Wiener, 1948 Panteli et al., 2017	Albert et al., 2000 Gao et al., 2016	Bruneau et al., 2003 Francis and Bekera, 2014 Panteli et al., 2017	Rane et al., 2024	Pariès and Wreathall, 2017		Holling, 1996 Francis and Bekera, 2014 Ivanov, 2020
Ecology & Social Systems		Holling, 1973 Hashimoto et al., 1982 Ives, 1995	Bodin and Crona, 2009	Ives, 1995 Sundstrom et al., 2014	Maillart et al., 2024	Berkes et al., 1998 Lebel et al., 2006 Mahajan et al., 2022	Maillart et al., 2024	Folke et al., 2010 Leslie and McCabe, 2013 Francis and Bekera, 2014 Allen et al., 2018
Climate Change & Disaster Management		Dawson et al., 2011 Panteli et al., 2017 Aydin et al., 2017	Jacobs and Cramer, 2017 Aydin et al., 2017	Bruneau et al., 2003	Arnold et al., 2004 Aerts et al., 2018	Dawson et al., 2011 Mahajan et al., 2022	Moreno and Shaw, 2018	Alexander, 2013 Meerow et al., 2016 Adger et al., 2005 Leichenko and O’Brien, 2024
Psychology & Neuroscience		Manikis et al., 2019 Ungar, 2021		Mcewen, 2004 Rakesh et al., 2019	Klein, 2008 Bada et al., 2019 Kalisch et al., 2024	Bada et al., 2019	Rutter, 1987 Masten and Osofsky, 2010	Wagnild and Young, 1993 Luthar et al., 2000 Campbell-Sills and Stein, 2007 Feder et al., 2019
Public Health		Brauer, 2008 Luke and Stamatakis, 2012	Luke and Stamatakis, 2012	Cutter et al., 2010 Kruk et al., 2017	Arnold et al., 2004	Barasa et al., 2018 Haldane et al., 2021	Gibbs et al., 2013 Aase et al., 2020	Norris et al., 2007 Barasa et al., 2018
Management	Organizational			Kuypers and Maillart, 2018	Klein, 2008 Bada et al., 2019 Almeida et al., 2020	Lanzara, 1999 Westrum, 2004 Weick and Sutcliffe, 2011 Henfridsson and Bygstad, 2013 Barasa et al., 2018 Bada et al., 2019 Boh et al., 2023		Westrum, 2004 Teece, 2007 Vogus and Sutcliffe, 2007 Bhamra et al., 2011 Burnard and Bhamra, 2011 Lengnick-Hall et al., 2011 Boin and van Eeten, 2013 Henfridsson and Bygstad, 2013 Skopik et al., 2016 Dupont, 2019 Barasa et al., 2018 Duchek, 2019 Calder, 2021 Hepfer and Lawrence, 2022 He et al., 2022
	Risk & Cyber		Gillard et al., 2023	Linkov et al., 2013 Panteli et al., 2017	Aerts et al., 2018 Almeida et al., 2020	Lostri et al., 2022 Boh et al., 2023	Kuypers and Maillart, 2018	Basel Committee on Banking Supervision, 2011 Björck et al., 2015 Woods, 2015 Skopik et al., 2016 Linkov and Kott, 2018 Dupont, 2019 Hausken, 2020 Calder, 2021 Lostri et al., 2022

Table 2: **Classification of Resilience-related Publications.** This table provides the classification of 89 relevant, impactful and supporting publications associated with resilience according to their corresponding topic/-s and used methodology/-ies. This table allows to extract the key research gaps, where no publication was found (red cells) or only very few (orange cell). Topic-methodology intersections in grey show other research that are less relevant for this research.

Third, **methodological fragmentation and limited transfer**. This gap becomes especially visible in the cross-disciplinary classification of the 89 resilience-related publications summarized in Table 2. Simulation models, network analyses, and quantitative resilience metrics are well established in engineering, complex systems, and climate research (Francis & Bekera, 2014). Their systematic integration into domains central to organizational and managerial resilience, such as cybersecurity, risk management, and strategic governance, remains much more limited. Table 2 shows that *Markers & Metrics* and *Technology* are comparatively underrepresented in these fields, and that simulation-based approaches are rarely incorporated into organization-level governance frameworks (Kuypers & Maillart, 2018; Linkov et al., 2013). The result is continued reliance on maturity models, expert judgment, or audit documentation: approaches that may support compliance, but rarely capture adaptive performance under real-world conditions.

Even within cybersecurity, formal models of attack–defense dynamics exist (Aydin et al., 2017), yet they are seldom integrated into broader organizational resilience architectures. Other transferable approaches, including ecological network analysis (Luke & Stamatakis, 2012), transportation recovery modeling (Jacobs & Cramer, 2017), and agent-based infrastructure risk models (Dawson et al., 2011), also remain underused in digital resilience planning. The problem is therefore not methodological scarcity. It is the weak translation of technical advances in resilience measurement into governance tools that organizations can actually use (Aerts et al., 2018; Heinimann & Hatfield, 2017).

These gaps do not remain separate. They reinforce one another. Conceptual plurality complicates measurement. Limited empirical validation weakens cumulative evidence.

Methodological fragmentation restricts the transfer of analytical tools across domains. As a result, resilience research contains substantial theoretical and methodological resources, but still lacks sufficient integration to support cumulative, organization-ready application. Without standardized, validated, and scalable metrics embedded in governance structures, resilience is likely to remain more a normative aspiration than a systematically cultivated capability.

Mathematical approaches from dynamical systems theory, such as basin stability, return dynamics, and perturbation–response analysis, provide promising foundations for formally characterizing resilience and tracing system adaptation over time (Krakovská et al., 2023). As organizational processes increasingly depend on interconnected digital infrastructures, data flows, and platform-based ecosystems, these approaches offer a useful basis for modeling how adaptive capacity emerges and evolves within digitally mediated organizational systems. 

3.1 From Organizational Resilience to Digital Organizational Resilience

Organizational resilience has traditionally been understood as the capacity of firms to absorb shocks, adapt to disruption, and sustain core functions over time (Burnard & Bhamra, 2011; Duchek, 2019). Earlier work emphasized managerial capabilities, organizational learning, and structural flexibility as central mechanisms through which organizations respond to uncertainty and unexpected disturbance. Contemporary organizations, however, increasingly operate through digitally mediated processes, interconnected infrastructures, and platform-based ecosystems (Gillard et al., 2023; Hepfer & Lawrence, 2022). As core activities become embedded in information systems, adaptive capacity is shaped not only

by organizational structures and managerial decisions, but also by technological architectures, data flows, and algorithmic coordination mechanisms (Liu et al., 2022).

In such environments, resilience emerges through the interaction between organizational actors and the digital infrastructures through which organizational processes are carried out. Disruptions therefore propagate not only through formal structures and routines, but also through technological dependencies, inter-organizational networks, and platform ecosystems. Understanding resilience in contemporary organizations consequently requires attention to how adaptive responses are enabled, constrained, and coordinated through digital systems.

We therefore conceptualize *digital organizational resilience* as the socio-technical specification of organizational resilience in digitally mediated environments (Duchek, 2019; Hepfer & Lawrence, 2022). This is not a separate theoretical domain. It clarifies how organizational resilience is enacted when adaptive capacity is distributed across digital infrastructures, institutional arrangements, and inter-organizational ecosystems (Janssen et al., 2006; Liu et al., 2022). In this view, resilience remains a dynamic capability (Duchek, 2019), but its realization increasingly depends on digitally embedded coordination mechanisms and infrastructural interdependencies (Boin & van Eeten, 2013; Linkov et al., 2013).

3.1.1 Digital Resilience Management Research

Once resilience is understood as a socio-technical capability emerging in digitally mediated organizational systems, the limits of the current literature become clearer. Research

on digital resilience has expanded, but it remains fragmented across several empirical streams. That fragmentation itself points to the need for a more integrative framework. Existing work spans cybersecurity, digital infrastructure governance, and information systems management, and generally emphasizes the role of technological architectures, organizational capabilities, and inter-organizational coordination mechanisms in shaping adaptive capacity. Even so, much of this work still addresses only partial aspects of resilience in digital environments.

~~A first~~ stream focuses on the resilience of digital infrastructures and technological systems. Studies in this area examine how system architectures, redundancy mechanisms, and recovery strategies enable digital infrastructures to withstand and recover from disruption. Research on cyber resilience, for example, emphasizes detection capabilities, adaptive response mechanisms, and system reconfiguration as means of reducing the impact of adversarial threats (Dupont, 2019; Linkov et al., 2013). Work on critical infrastructure protection similarly highlights the importance of network robustness, modularity, and redundancy in sustaining essential digital services during disturbance. This literature provides valuable insight into the technical foundations of resilience, but it often treats organizations mainly as operators of technological systems rather than as adaptive actors embedded in broader socio-technical environments.

~~A second~~ stream examines resilience from an organizational and managerial perspective. This work highlights the role of leadership, governance structures, and organizational learning in enabling firms to anticipate, respond to, and adapt to digital disruption. Research on digital risk management and cybersecurity governance emphasizes decision-

making processes, incident response capabilities, and cross-functional coordination mechanisms as important contributors to organizational resilience (Boin & van Eeten, 2013).

These approaches are useful, but they often focus primarily on internal capabilities and may understate the extent to which resilience depends on external technological infrastructures and inter-organizational dependencies.

~~A third stream~~ addresses resilience at the level of digital ecosystems and inter-organizational networks. Contemporary organizations increasingly operate within digital environments composed of interconnected platforms, supply chains, and information-sharing communities. Research on networked governance and digital collaboration shows that resilience in these settings depends not only on the capabilities of individual organizations, but also on the collective capacity of actors to share information, coordinate response, and adapt to evolving disruption (Janssen et al., 2006; Liu et al., 2022). Cybersecurity information-sharing initiatives, platform governance mechanisms, and collaborative incident-response communities illustrate how resilience can emerge through coordinated action across organizational boundaries.

What remains needed is a perspective that links resilience mechanisms to the socio-technical environments in which they operate. In digitally mediated organizational systems, resilience depends not only on internal organizational capabilities but also on the infrastructures and ecosystems through which organizational processes are enacted. The following section therefore introduces the Enactment-Scope framework, which links mechanisms of resilience enactment to the scopes of coordination within which they operate.

4 THE ENACTMENT-SCOPE FRAMEWORK

The preceding analysis points to a basic tension in resilience research. Resilience is widely invoked as a desirable organizational capability, yet much of the literature still isolates only partial elements of it – infrastructure robustness, organizational preparedness, or inter-organizational coordination – without showing clearly how these elements combine to produce adaptive performance in digitally mediated environments (Boin & van Eeten, 2013; Burnard & Bhamra, 2011). As a result, resilience is often discussed as a broad aspiration, yet remains difficult to operationalize, compare, or evaluate across organizational settings (Ibne Hossain et al., 2020; Linkov et al., 2013).

The more challenging task, then, is not simply to argue that resilience matters, but to demonstrate specific methods by which it becomes an observable and governable organizational capability. Figure 1 illustrates this move from strategic aspiration to operationalization. Building on that logic, the Enactment-Scope framework specifies where resilience mechanisms operate and how they are realized across digitally mediated organizational systems. Operational indicators make adaptive mechanisms visible. Design principles define decision rights and structure coordination across organizational units. Benchmarks and warning systems align incentives and signal movement away from desired states. Together, these mechanisms translate resilience from rhetorical commitment into enacted organizational capability in practice (Boin & van Eeten, 2013; Ibne Hossain et al., 2020; Linkov et al., 2013).

That still leaves a question: where exactly do these mechanisms operate in digitally mediated organizational systems? Contemporary organizations function within socio-

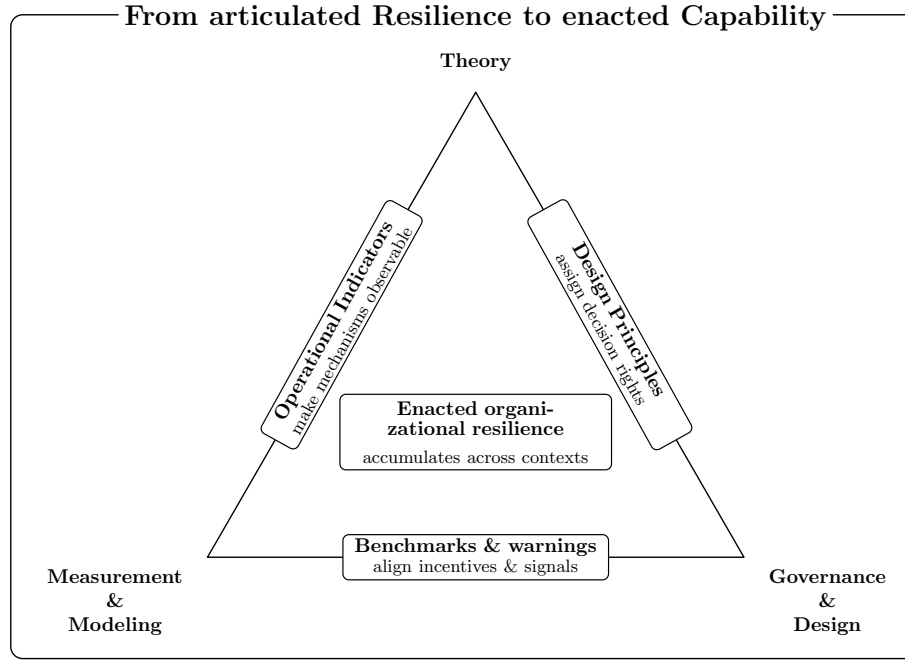


Figure 1: **From articulated resilience to enacted capability.** Operational indicators make adaptive mechanisms observable. Design principles structure decision rights and coordination. Benchmarks and warning systems align incentives and signal deviation from desired states. Together, these mechanisms help translate resilience from rhetorical commitment into enacted organizational capability.

technical environments composed of digital infrastructures, organizational processes, and inter-organizational ecosystems (Gillard et al., 2023; Liu et al., 2022). Resilience mechanisms operate across different socio-technical arrangements and scopes of coordination. The Enactment-Scope framework is crafted to seize that structure by organizing resilience mechanisms according to how they are enacted and the scopes within which they operate.

4.1 Enactment of Resilience

The first dimension of the framework concerns the *enactment* of resilience. Resilience is not treated as an abstract attribute that organizations simply possess. It emerges through concrete mechanisms embedded in technological systems, organizational practices, and governance arrangements (Burnard & Bhamra, 2011; Duchek, 2019). These mechanisms include technological safeguards like redundancy and automated recovery, organizational

practices like incident-response routines and learning processes, and collaborative arrangements like information sharing and coordinated response across organizations (Dupont, 2019; Gillard et al., 2023; Linkov et al., 2013).

Viewing resilience through enactment shifts attention from abstract claims to concrete socio-technical activity. Adaptive capacity arises from interactions among human actors, digital infrastructures, and institutional arrangements (Dupont, 2019; Liu et al., 2022). In digitally mediated environments, resilient performance depends on whether mechanisms are implemented, activated under disruption, and enable organizations to detect, respond to, and adapt as conditions evolve (Duchek, 2019; Linkov et al., 2013). This makes enactment analytically important as it shifts focus from theoretical claims to practical implementations, thereby providing a clearer understanding of resilience in action.

4.2 Scope of Resilience

The second dimension concerns the *scope* at which resilience mechanisms operate. Contemporary organizations rarely function in isolation, embedded in digitally interconnected environments composed of infrastructures, platforms, and collaborative networks (Gillard et al., 2023; Liu et al., 2022). Resilience mechanisms may operate across multiple scopes of coordination rather than within a single organizational boundary.

At the *organizational* scope, resilience mechanisms are primarily internal to the firm, including governance structures, risk-management practices, and technological safeguards intended to maintain continuity of operations (Burnard & Bhamra, 2011; Duchek, 2019). At the *infrastructural* scope, resilience depends on the robustness and recoverability of

shared digital systems like communication networks, cloud platforms, and critical information infrastructures (Ibne Hossain et al., 2020; Linkov et al., 2013). At the *ecosystem* scope, resilience emerges through coordinated action across organizational boundaries, including information sharing, collaborative incident response, and collective governance arrangements (Dupont, 2019; Gillard et al., 2023).

Distinguishing these scopes clarifies that resilience in digital environments often depends on interactions among mechanisms operating across several layers of coordination (Boin & van Eeten, 2013; Ibne Hossain et al., 2020). An organization's response to a cyber incident may depend not only on its internal detection and response capabilities, but also on the resilience of external infrastructures and the responsiveness of partner organizations across the wider ecosystem (Dupont, 2019; Gillard et al., 2023). Scope matters because the location of a resilience mechanism conditions both what it can achieve and what it depends on, highlighting the need for understanding interactions across different scopes.

4.3 Integrating Enactment and Scope

Bringing enactment and scope together provides a lens for examining how resilience mechanisms interact across socio-technical configurations. The framework shows how adaptive capacity emerges through the interaction of technological infrastructures, organizational practices, and ecosystem-level coordination mechanisms (Boin & van Eeten, 2013; Duchek, 2019; Liu et al., 2022). It makes possible analysis of how resilience mechanisms reinforce one another, and where they generate dependencies, bottlenecks, blind spots, or misalignments across coordination scopes within digitally mediated organizational systems.

This is also where the framework departs from more generic multi-level accounts. It does not simply note that resilience involves several layers. It clarifies why resilience may appear strong at one scope while remaining fragile at another. An organization may possess well-developed internal response routines but depend on brittle infrastructures or unresponsive ecosystem partners. Conversely, robust infrastructures may not produce resilience when organizational routines are poorly aligned or cross-boundary coordination fails. The framework makes visible not only the distribution of resilience mechanisms but also the interdependencies through which resilient or non-resilient outcomes are produced.

The framework also helps explain why resilience is often difficult to operationalize in organizational contexts, primarily due to the complexity of integrating mechanisms across various socio-technical layers. Many resilience initiatives focus on mechanisms at only one scope – internal preparedness, technological redundancy, or external coordination – while neglecting how these mechanisms interact across socio-technical layers (Boin & van Eeten, 2013; Burnard & Bhamra, 2011; Linkov et al., 2013). By linking mechanisms of enactment to the scopes within which they operate, the Enactment–Scope framework offers a more structured way to analyze how resilience arises from the interplay of organizational, technological, and ecosystem-level processes.

In that sense, the framework advances resilience research in two ways. The framework shows how adaptive capacity emerges from the interaction of technological infrastructures, organizational practices, and ecosystem-level coordination mechanisms (Boin & van Eeten, 2013; Duchek, 2019; Liu et al., 2022). Second, it provides an analytical

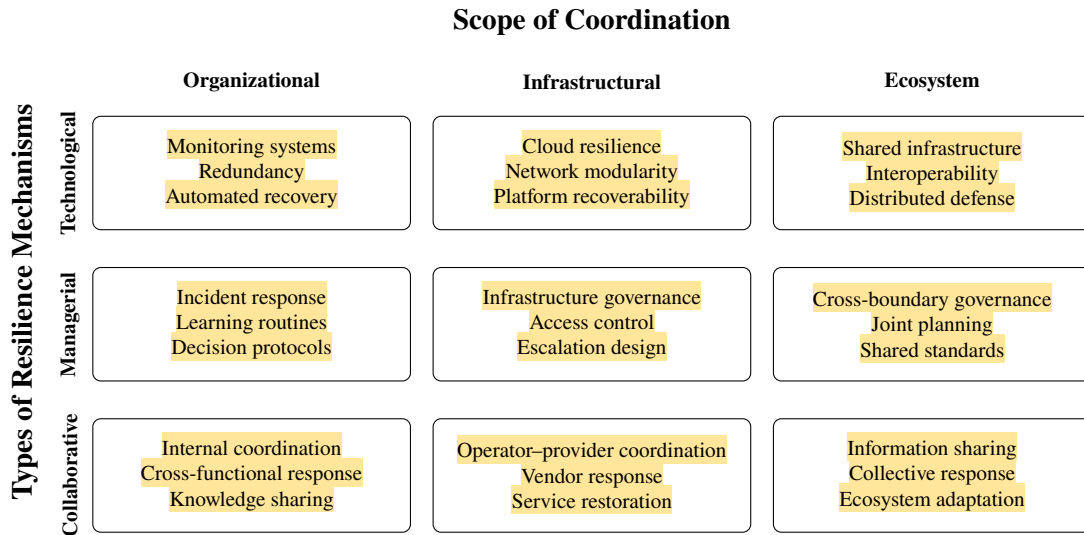


Figure 2: **The Enactment–Scope Framework of Digital Organizational Resilience.** Resilience mechanisms are enacted through technological, managerial, and collaborative processes, and operate across different scopes of coordination including organizational, infrastructural, and ecosystem levels.

basis for developing operational measures that capture how resilience mechanisms function across digitally mediated organizational environments, thereby supporting empirical assessment, simulation-based analysis, and governance-oriented evaluation of digital resilience (Ibne Hossain et al., 2020; Linkov et al., 2013). Rather than treating resilience as an undifferentiated organizational virtue, the framework makes it possible to specify where adaptive capacity is located, how it is enacted, and why it succeeds or fails under conditions of disruption.



5 RESEARCH DIRECTIONS FOR DIGITAL ORGANIZATIONAL RESILIENCE

The Enactment–Scope framework is useful only if it generates concrete research questions. This section develops that next step. The focus here is digital organizational resilience, and especially cyber resilience, because digital environments make the core analytical problems unusually visible. In such settings, resilience cannot be treated as a slogan or

as a compliance label attached after the fact. It has to be studied as a socio-technical capability that is built, enacted, and tested under conditions of persistent disruption.

This shift changes the object of inquiry. Once resilience is treated as something that can be observed in behavior, infrastructure, and governance arrangements, it becomes possible to evaluate it rather than merely invoke it. Once it is treated as cumulative rather than static, questions of learning, memory, reinforcement, and erosion move to the center. And once it is situated in digitally mediated environments, adaptive capacity can no longer be reduced to managerial discretion or organizational culture alone. It also depends on architectures, platforms, institutional settings, and the ways these elements condition what forms of response are possible.

The research agenda that follows is organized around two linked questions. The first concerns *enactment*: through what mechanisms does resilience move from stated intention to operational capability? The second concerns *scope*: at what organizational and institutional levels do these mechanisms actually operate? These questions follow directly from the earlier diagnosis of resilience research. Resilience remains rhetorical when organizations lack observable indicators, governance structures that allocate decision authority, and evaluative benchmarks that align action with adaptive goals. At the same time, resilience in digitally mediated environments is distributed. It is not located in one team, one platform, or one formal rule. It is produced across routines, infrastructures, institutional arrangements, and inter-organizational ecosystems.

This is also where information systems research has a distinctive advantage. The

field already addresses digital infrastructures, platform governance, socio-technical coordination, and cross-boundary information flows. These are not peripheral concerns here. They are the substance of the problem. The sections below therefore develop three research directions: operationalizing resilience enactment, examining resilience across organizational and institutional scopes, and tracing how resilience evolves over time.

5.1 Operationalizing Resilience Enactment

A first priority is to make resilience observable in practice. That is the central task of the enactment dimension of the framework. Organizations frequently invoke resilience in strategic language, governance documents, and policy discourse, yet it often remains unclear how resilience should actually be identified during disruption. IS research is well placed to address that problem because it can connect behavioral indicators, computational traces, and organizational processes in ways that make adaptive capacity empirically visible.

The challenge is to move beyond metaphor. It is no longer enough to say that resilience matters. Research has to specify how resilience appears in practice and how it can be distinguished from symbolic claims about preparedness or adaptability. Comparative work across domains already suggests that resilience emerges through the interaction of routines, infrastructures, and institutional processes (Manikis et al., 2019; Pumpuni-Lenss et al., 2017). What is still missing is a more systematic basis for turning those observations into measures and design principles that organizations can actually use.

Figure 1 is intended to support that move. It connects theory, measurement, and gov-

ernance as the three domains through which resilience becomes enacted capability. Each side of the triangle captures a different mechanism of translation. *Operational indicators* make resilience observable rather than merely symbolic. *Design principles* allocate decision rights and structure adaptive responses. *Benchmarks and warnings* align incentives while producing actionable signals. The figure matters less as a visual device than as an analytical claim: resilience becomes governable only when these domains are linked rather than treated separately.

One promising line of work concerns the development of behavioral and computational indicators that make resilience empirically assessable (Gillard et al., 2023; Kuypers & Maillart, 2018). Indicators such as coordination speed, the ability to reconfigure infrastructures, and the variability of response under pressure can help turn resilience from an aspirational concept into a measurable one (Mahajan et al., 2022). They also make comparison possible. Organizations can benchmark themselves, and researchers can begin to accumulate evidence across cases. Yet the development of such indicators is not simply a technical exercise. It also depends on a theoretical account of how behaviors at individual and collective levels scale into system-level outcomes (Luthar et al., 2000).

Digital environments make this task especially urgent. Disruptions unfold quickly, interactions are distributed, and many adaptive signals are embedded in digital traces rather than retrospective narratives. Work in behavioral analytics and AI-assisted modeling therefore offers a useful path forward, particularly where resilience indicators can be calibrated against observable signatures in digital artifacts (Rane et al., 2024).

Computational approaches, including agent-based modeling and simulation, are especially useful here. They allow researchers to explore how coordination patterns, incentive structures, and infrastructural configurations shape the capacity of organizations to absorb, reconfigure, and respond to disruption (Arnold et al., 2004; Gillard et al., 2023). Used carefully, such methods do more than illustrate scenarios. They allow theory to be refined against simulated and empirical evidence and help bridge the gap between conceptual language and organization-level practice. In that sense, operationalization is not only about measurement. It is also about making resilience a capability that can be examined, compared, and deliberately improved.



5.2 Resilience Across Organizational and Institutional Scopes

A second research direction develops the scope dimension of the Enactment–Scope framework by asking where resilience mechanisms operate and how they are distributed across organizational, infrastructural, institutional, and inter-organizational settings.

In digitally mediated environments, adaptive capacity is inherently distributed. It does not sit neatly inside a single team, technology, or governance layer. Governance architectures can enable resilience, but they can also constrain it by creating ambiguity in decision rights, accountability gaps, and escalation failures (Lebel et al., 2006; Ostrom, 1990). These frictions may remain latent in stable periods, but they become visible under stress. They appear as delayed responses, constrained options, unclear ownership, and coordination breakdowns.

Studying these dynamics requires more than abstract references to governance. It calls

for approaches that connect institutional theory, regulatory ethnography, and actor-network perspectives with empirical work on digital infrastructures and everyday organizational practice. The point is to understand how resilience is enabled or blocked in the lived realities of digitally coordinated work, not merely in formal schemes of control.

Figure 3 introduces an opportunity matrix that maps research directions for digital organizational resilience across four scopes: *micro-level routines*, *meso-level infrastructures*, *macro-level institutions*, and *interorganizational ecosystems*. This matrix helps identify where resilience mechanisms can be studied and where they may require institutional support. It also reveals a pattern that matters. Institutional friction and cooperation problems become especially salient at meso and macro levels, while integrative opportunities run across all four.

This matrix does not replace the enactment perspective. It complements it. The enactment dimension clarifies *how* resilience becomes operationalized through indicators, design principles, and evaluative benchmarks. The scope dimension clarifies *where* those mechanisms matter. Together, they provide the central logic of the Enactment–Scope framework by linking enacted mechanisms to the distributed locations in which adaptive capacity is organized.

A related issue concerns cooperation under pressure. Time-critical crises rarely respect organizational boundaries. Response depends on whether actors can coordinate quickly across firms, infrastructures, and institutional settings. Yet this is exactly where trust asymmetries, activation delays, and incentive misalignments often undermine per-

Directions	Routines (micro)	Infrastructures (meso)	Institutions (macro)	Ecosystems (inter-org)
Operational indicators	●	●	●	●
Integrative frameworks	○	●	●	●
Comparative & cumulative	○	●	●	○
Temporal/process markers	●	○	●	●
Normative/value-based	○	●	●	●
Governance and incentives	○	●	●	●

Legend: ● high opportunity; ● promising ; ○ low opportunity.

Table 3: Opportunity matrix locating research directions across organizational scopes (micro routines, meso infrastructures, macro institutions, inter-organizational ecosystems).

formance. Theories of collective action, modular governance, and platform-mediated collaboration therefore provide an important starting point, especially in digital threat environments where rapid mobilization is essential (Brauer, 2008; Mahajan et al., 2022; Ostrom, 1990).

Empirical work on modular security platforms such as MISP is especially relevant here because it shows how structural design can enable or inhibit timely collaboration. Participatory simulations and controlled experiments add another layer by identifying thresholds for cooperative engagement and the points at which collaboration begins to fail structurally (Gillard et al., 2023; Skopik et al., 2016). Disaster-response research offers additional design principles for cross-organizational information sharing through IT systems, and many of those insights are transferable to adversarial digital environments (Arnold et al., 2004). Whether resilience becomes a substantive capability or remains largely symbolic depends in no small part on how these institutional arrangements and cooperation dynamics are configured across scopes.

5.3 Temporal Dynamics of Resilience Enactment

A third research direction extends the enactment perspective by asking how resilience is formed and re-formed over time. The point is not simply to add a temporal layer to the studies discussed above, but to examine how enacted capacities are built, stabilized, weakened, or redirected through experience, feedback, and institutional choice. On this view, resilience is neither a fixed organizational attribute nor something secured by resource availability alone. It is a contingent and relational capacity whose durability depends on whether earlier adaptations are reinforced, remembered, and carried forward, or instead neglected and allowed to erode (Luthar et al., 2000; Rutter, 1987). Yet this temporal dimension remains underdeveloped in much of the IS literature, which still tends to focus on discrete events or isolated responses rather than longer trajectories of adjustment. Organizations do not confront each disruption from a blank slate. They carry residues of earlier crises: routines, assumptions, temporary fixes, and interpretive habits that may support later adaptation, lock them into rigid responses, or simply fade over time. Any persuasive account of resilience therefore has to explain not only how organizations respond in the moment, but also how adaptive capacity is retained, revised, or lost across successive episodes of disruption.

Useful foundations for this line of inquiry already exist in systems theory and ecology. Holling's work on adaptive cycles, for example, draws attention to recurring phases of growth, conservation, release, and reorganization, making clear that resilience is tied to sequences and transitions rather than to any single moment of adjustment (Holling, 1973). Wiener's cybernetics places feedback at the center and shows how systems may be stabilized through corrective feedback or pushed toward instability through reinforcing

feedback (Wiener, 1948). Taken together, these traditions suggest that resilience depends not only on an organization's immediate ability to cope with disruption, but also on whether effective responses are reinforced, whether lessons become institutionalized, and whether failure produces revision rather than repetition. Under some conditions resilience accumulates over time; under others it decays through forgetting, inertia, or maladaptation.

Studying these dynamics calls for methodological pluralism. Longitudinal case studies are needed to follow how organizations adapt across repeated disruptions rather than single events. Process tracing can then identify the mechanisms through which learning practices, governance arrangements, and changes in technical systems shape later outcomes. Computational models offer complementary leverage by making it possible to examine how different rates of reinforcement, forgetting, or knowledge transfer generate different long-run resilience trajectories. ~~Taken together, these approaches would allow IS research to move beyond largely static conceptions of resilience and instead analyze it as a socio-technical capability that can strengthen, deteriorate, or mutate as organizations are repeatedly disturbed.~~

5.4 Toward an Integrative Enactment-Scope Research Agenda

Taken together, these research directions point toward a broader agenda: resilience needs to be studied by linking mechanisms of enactment to the organizational scopes within which they operate. Fragmented perspectives will only take the field so far. What is needed instead is a socio-technical account that can relate behavioral processes, institutional conditions, and temporal development without collapsing them into a single level of analysis. This ambition is consistent with systems and ecological traditions that treat resilience as

an emergent effect of feedback, interdependence, and adaptation (Holling, 1973; Wiener, 1948). For IS research, that means building explanations that connect micro-level action, meso-level organizational and infrastructural dynamics, and macro-level governance constraints. It also means developing indicators that do more than register compliance. Useful indicators would need to capture adaptation as it unfolds: changes in response variability, shifts in coordination under stress, and the reconfiguration of technical and organizational arrangements during disruption (Aerts et al., 2018). Unless resilience is anchored in observable behavior, it risks remaining rhetorically attractive without becoming analytically precise.

At the same time, resilience does not emerge in a vacuum. The institutional and organizational environments in which actors operate shape what forms of adaptation are feasible in the first place. Rules, incentives, authority structures, and escalation pathways do not simply constrain action after disruption; they structure the range of possible responses before disruption occurs (Ostrom, 1990; Ungar, 2021). Existing field-based and computational studies already show that these conditions matter, often appearing as friction, delay, coordination breakdown, or reduced strategic flexibility. The harder task now is to turn these scattered insights into analytical frameworks that can travel across settings without losing explanatory force.

Figure 3 makes this research space visible by locating opportunities across micro behaviors, meso infrastructures, macro institutions, and inter-organizational ecosystems. Read together, Figures 1 and 3 define the central logic of the Enactment–Scope framework. The triangle specifies how resilience moves from rhetorical commitment to enacted

practice. The opportunity matrix shows where those mechanisms can be studied and where intervention is most likely to matter. Advancing this agenda will require research programs that combine theoretical ambition with methodological range, including computational modeling, longitudinal case analysis, and field-based validation. Used together, these approaches offer a more credible path toward cumulative and transferable knowledge about organizational resilience (Gillard et al., 2023; Kalisch et al., 2024; Teece, 2007). More importantly, they provide a stronger basis for IS theorizing about digital resilience by showing how adaptive capacity emerges through the interaction of technological architectures, organizational practices, and institutional environments. On this view, resilience is neither a slogan nor a checklist. It is an enacted, distributed, and revisable socio-technical capability for maintaining and reorganizing coordinated action under persistent uncertainty (Weick & Sutcliffe, 2011; Woods, 2015).

6 IMPLICATIONS FOR RESEARCH & PRACTICE

The three gaps identified in this review – conceptual ambiguity, limited empirical grounding, and methodological fragmentation – do not point to isolated weaknesses. Together, they highlight a wider issue with the way resilience is studied in digitally mediated environments. The issue is not just that the literature is incomplete. Rather, resilience is still too often invoked in general terms, measured inadequately, and only operationalised after disruption has already occurred. The following implications take a different view. They treat resilience as a socio-technical capability that must be defined more precisely, observed in action and linked to effective interventions.

Firstly, resilience should not be considered a fixed organisational attribute. It is not

merely a residual label attached to organisations that appear to cope well after disruption, either. It is better understood as an emergent socio-technical capability. This is important because the literature often moves too easily between resilience, robustness, recovery and agility (Duchek, 2019; Woods, 2015). The overlap is real, but so are the differences. In digital settings especially, resilience is not confined to one asset, one team, or one managerial disposition. Rather, it emerges through interaction between people, technical systems, routines, escalation pathways and governance arrangements. A good example is cyber incident response, where it isn't about one particular resource but rather how well the analysts, security software, and other reporting protocols, along with other channels for information sharing, can collaborate successfully within time constraints. Resilience framed in such a manner will give IS researchers a better starting point for their analyses, while facilitating a more productive interaction between IS research and dynamic capability studies. (Boh et al., 2023; Henfridsson & Bygstad, 2013), while remaining compatible with systems and ecological traditions (Holling, 1973; Wiener, 1948).

A second implication follows from this almost immediately. Resilience must be made observable. Otherwise, it remains just a slogan. Behavioural and computational indicators are important not because they make resilience appear more scientific, but because they make the concept empirically accountable. Plausible candidates include coordination speed, stakeholder support, community ties, the capacity to reconfigure infrastructures, and variation in response under stress (Gillard et al., 2023; Mayer & Roche, 2021; Pariès & Wreathall, 2017). Once specified, such indicators can be studied through agent-based simulation, network analysis, or probabilistic modelling (Helbing, 2012). This is especially important in digital spaces, where adaptation is spread out, needs to happen quickly, and

is often only partially visible. A simulation model can show how changes in how quickly information is shared or how weak incentives for collaboration can change how people work together during cyber incidents. That is the point. When you can see something, you can compare resilience. It can also be questioned and changed.

This also means we spend less time treating compliance artifacts as evidence of resilience. Although audits, certifications, and maturity assessments are not useless, they are weak indicators of adaptive capacity in practice. (Skopik et al., 2016). They inform us of the organizations that claim to be prepared to do so. They divulge scant information regarding the ramifications of infrastructure failure, the dissolution of coordination, or the transformation of circumstances midway. Field-based work is therefore essential. Longitudinal case studies, trace data, and ethnographic approaches are important here. Research in healthcare and public health already shows that resilience can be evident in everyday practice while remaining only partially institutionalized (Barasa et al., 2018; Kruk et al., 2017). The same opportunity exists in digital infrastructures, where incidents and near-misses reveal how organisations improvise, adapt, stall or fail over time. In some cases, disruption may trigger processes of renewal reminiscent of Schumpeter's 'creative destruction', in which routines or infrastructures that no longer fit prevailing conditions are replaced by more adaptive ones (Schumpeter, 1942). This should not be romanticised. However, it does reinforce the broader point that resilience becomes analytically meaningful when studied in action rather than inferred from documentation.

Another implication is that adaptive capacity does not belong to only one level of analysis. Rather, it unfolds across micro-level routines, meso-level infrastructures, macro-level

institutions, and inter-organizational ecosystems. This is one reason why single-level explanations often feel inadequate. A cyberattack may begin with analyst decisions at the micro level, spread through infrastructural dependencies at the meso level, and then be redirected or constrained by regulation, liability, or cross-organizational coordination at broader scales. The mechanisms are distributed. IS research is unusually well placed to deal with that because it can combine process tracing, computational modeling, and comparative case analysis without forcing the problem into a single analytical register (Folke et al., 2010; Ostrom, 1990; Vogus & Sutcliffe, 2007). That is a real advantage of the field. It also creates room for indicators – such as coordination speed, recovery sequence, or adaptive reconfiguration – that travel across contexts without collapsing into empty abstraction.

There is also a practical implication that should be stated plainly. Better concepts and indicators won't matter much if they don't influence how organizations prepare, make decisions, and coordinate. To that end, the field of resilience research must evolve to become more managerially useful without succumbing to managerial clichés. This requires design principles, benchmarks, early warning systems, and governance mechanisms that serve a purpose beyond mere decoration of annual reports (Calder, 2021; Lengnick-Hall et al., 2011; Weick & Sutcliffe, 2011). The risk is obvious: Organizations can demonstrate resilience long before they operationalize it. Metrics theater is not rare. Dashboards exist, escalation paths are documented, and the language of resilience is prevalent. Yet, adaptive capacity remains thin when pressure actually arrives. Therefore, linking measurement to governance is not optional. In practice, this may involve real-time monitoring of the adaptive response, escalation protocols that clarify authority during disruption, or governance arrangements that assign coordination responsibilities before a crisis begins.

When considered as a whole, these findings suggest a need for a more rigorous research agenda in the field of IS. The field does not need more generic endorsements of resilience or another round of compliance-friendly abstraction. To achieve this, it is essential to develop more sophisticated concepts, strengthen the empirical foundation, and devise methods that can elucidate the process by which adaptive capacity emerges in the face of ongoing disruption. For scholars, that creates room for more cumulative theory. It also creates room for better evidence. The conversation is shifted for practitioners from symbolic commitment to capabilities that can be observed, tested, and revised. Reframed in these terms, resilience becomes less a rhetorical aspiration than a grounded socio-technical capacity for operating under strain in digitally mediated environments.

7 CONCLUSION

Resilience, a buzzword in management and information systems research, often falls short of its promise, remaining more proclaimed than enacted. Organizations use it widely, but in practice, they face persistent ambiguity, methodological limitations, and weak empirical grounding. This ambiguity frequently translates into compliance-oriented practices that signal legitimacy without cultivating genuine adaptive capability.



This review identified three core limitations in the current literature. First, construct clarity is lacking: resilience is often confused with robustness, agility, or redundancy, which obscures its unique mechanisms. Second, operationalization remains underdeveloped: organizations and scholars lack reliable indicators to capture adaptive performance during disruption. Third, cumulative evidence remains scarce: insights are fragmented

across disciplines and contexts, preventing generalizable knowledge from emerging.

To address these gaps, we propose a research agenda that conceptualizes resilience as a measurable socio-technical capability. Four directions are particularly important. First, operational indicators should render resilience observable through behavioral and computational markers. Second, integrative design principles must connect routines, infrastructures, and governance arrangements across organizational and technological layers. Third, benchmarks and early-warning systems are needed to align incentives and prevent the emergence of “metrics theater.” Finally, comparative and cumulative empirical research – combining simulation, computational modeling, and field data – can help establish transferable insights across organizational settings and levels of analysis.

By following these directions, resilience is repositioned from a rhetorical aspiration to an enacted socio-technical capability. By advancing conceptual clarity, measurable indicators, and cumulative empirical knowledge, information systems research is uniquely positioned to transform resilience from a symbolic label into a tractable capability for navigating disruption in complex digital environments. More broadly, this review positions resilience as an emergent socio-technical capability arising from interactions among digital infrastructures, organizational practices, and institutional arrangements. By clarifying these mechanisms and identifying avenues for empirical and computational investigation, this study provides a foundation for cumulative theory development on resilience in digitally mediated organizational systems.

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APPENDIX

A FOUNDATIONAL CONSTRUCTS AND DISCIPLINARY PERSPECTIVES ON RESILIENCE

This appendix supplements the background section by detailing key theoretical perspectives that have shaped resilience thinking across disciplines. Each subsection identifies the core logic, constructs, and limitations associated with a given paradigm, and highlights their implications for organizational implementation.

A.1 Mechanical and Homeostatic Origins

Resilience, a scientific concept born in the 17th century, initially referred to the rebound of solid bodies or the echo of sound. Over time, it evolved to encompass broader scientific and systemic interpretations, particularly in biology and engineering (Alexander, 2013). In its early sense, resilience described a material's ability to recover its original form after a force was applied, like a spring returning to shape after being compressed. This notion aligns closely with homeostasis, a principle central to biological and engineering systems in the 20th century. Homeostasis describes a system's ability to maintain a stable internal state in the face of external changes, typically through feedback loops. The human body, for example, maintains a constant temperature through sweating or shivering (Wiener, 1948). Here, resilience is defined by the ability to return to a fixed, desirable state after small perturbations. Early analytical approaches to resilience focused on stability and return, the idea that a resilient system could recover to its original equilibrium after being disturbed. In engineering, resilience was formalized through performance-based criteria. A widely adopted framework defines resilience as the speed of recovery from failure, complementing reliability (the frequency of successful performance) and vulnerability (the severity of failure) (Hashimoto et al., 1982). These three criteria form a foundational model for

evaluating system performance under stress, particularly in water resource and infrastructure domains. These three criteria form a core framework for assessing system behavior under stress, particularly in water resource and infrastructure domains. In mathematical modeling, this is often represented using differential equations such as $\frac{dx}{dt} = f(x, t)$, where the equilibrium point x^* satisfies $f(x^*, t) = 0$ (Strogatz, 2018). If a system perturbed from x^* eventually returns to it, it is considered stable and by early definitions, resilient. A simple example is a swinging pendulum that returns to its resting point. This kind of behavior is also at the heart of classical control theory, which emerged to design systems that resist disturbances through corrective action. A system governed by $\frac{dx}{dt} = Ax + Bu$ can be stabilized using a control input like $u = -Kx$ to guide it back to equilibrium (Åström & Murray, 2010). This is how a self-balancing robot stays upright when nudged—by continually correcting its position based on feedback. Materials science defines resilience by how much energy a material can absorb within the elastic region before deforming permanently. This is quantified by the modulus of resilience: $U_R = \frac{\sigma_y^2}{2E}$, where σ_y is the yield stress and E is the Young’s modulus (Callister Jr & Rethwisch, 2020). Materials like spring steel exhibit high resilience, demonstrating the concept’s applicability across disciplines, from engineering to materials science, where the ability to absorb energy and recover is crucial.

While these approaches are powerful, they assume systems are designed to stay the same: disturbances are temporary, structure is fixed, and return to a baseline is both possible and desirable. As a result, mechanical models risk mistaking system stability for adaptive capacity – reinforcing symbolic notions of resilience without enabling transformation.

A.2 Shifting Toward Adaptive Capacity

As our understanding of nature and society evolved so did the limitations of this view. Many systems – ecological, technological, economic, and public health systems – don't merely rebound. They adapt. They transform (Folke et al., 2010). They face nonlinear changes, irreversible shifts, and strategic interactions (Gunderson & Holling, 2002). A forest recovering after a wildfire may not return to its previous state; it may reorganize into a different but still functional ecosystem (Walker et al., 2004). A supply chain disrupted by a pandemic may adapt by diversifying suppliers rather than returning to the previous model (Ivanov, 2020). Similarly, health systems recovering from crises such as pandemics must not only restore functionality but build capacity for future stressors by integrating local knowledge, equitable access, and preventive strategies (Kruk et al., 2017; Norris et al., 2007). This recognition marked a conceptual shift in resilience thinking – from homeostasis to adaptive capacity (Folke et al., 2004). Instead of assuming a fixed point of stability, newer approaches emphasized how systems persist through change, including transformation (Walker et al., 2004). This shift also led to stochastic models that quantify resilience under random disturbances using return variance and autocorrelation (Ives, 1995). In social-ecological contexts, this includes the ability of communities and ecosystems to co-adapt, as in coastal regions where human and environmental systems jointly manage flood risk through hybrid solutions like restored wetlands and elevated infrastructure (Adger et al., 2005).

Unlike mechanical approaches, these perspectives treat change as inevitable and often desirable – framing resilience not as returning to normal, but as reorganizing in ways that enable persistence under new conditions.

A.3 Dynamic and Strategic Perspectives

Ecological systems, for example, may exhibit multiple equilibria. A shallow lake might shift from clear to turbid due to nutrient overload and fail to recover even when nutrient levels decline – a dynamic that early models could not capture (Holling, 1973). Resilience in this context is the system’s capacity to absorb disturbance without shifting into an undesirable regime. Mathematical approaches evolved accordingly. However, dynamic and strategic perspectives often emerge from different disciplinary traditions, each with its own underlying definition of resilience. Such variation has motivated the creation of integrative typologies that map these competing interpretations and highlight the need for both context-specific operationalization and effective application (Meerow et al., 2016).

In network theory, the focus is not on a specific equilibrium but on maintaining core functionality amid partial failure. The Giant Component in a network ensures global connectivity. Its persistence depends on the average node degree $\langle k \rangle$ exceeding a critical threshold $p_c = 1/\langle k \rangle$ (Newman, 2018). This is crucial for systems like power grids, where a network’s ability to transmit electricity depends on keeping critical nodes connected even during outages (Peng et al., 2015). In ecology, network analysis has been used to model flows of energy and resources, assess systemic robustness, and identify critical structural dependencies, offering methodological parallels for resilience assessment in socio-technical systems (Luke & Stamatakis, 2012). In probabilistic models, resilience is expressed in terms of survival and reliability over time. The survival function $S(t) = \mathcal{P}(T > t) = 1 - F(t)$ indicates the probability that a component or system survives beyond time t (Walpole, 1998). This perspective is essential in contexts like infrastructure and public health, where resilience involves withstanding and recovering

from rare, high-impact events. For example, community resilience in public health relies on shared social capital, redundancy in care delivery, and early-warning information systems (Aldrich & Meyer, 2014; Haldane et al., 2021). As systems became more interconnected and dynamic, resilience analysis incorporated anticipation, adaptation, and decision-making under uncertainty. Formal system-level metrics have also been proposed to capture recovery dynamics and basin stability in complex systems (Krakovská et al., 2023). In modern control systems, such as model predictive control (MPC), a system forecasts future conditions and adjusts its behavior proactively (Almassalkhi & Hiskens, 2015). A microgrid using MPC can reroute energy to maintain voltage and frequency even during external shocks. Strategic behavior also emerged in game-theoretic models, which describe resilience as the result of interactions among rational agents. A cyber-physical system under threat may use game-theoretic reasoning to adaptively defend itself, optimizing control strategies in real time while anticipating potential attacks (Myerson, 2013; Zhu & Basar, 2015). Likewise, adaptive governance in climate resilience incorporates participatory approaches and local feedback to design flexible policy mechanisms (Lebel et al., 2006). Social-ecological perspectives in dynamic contexts highlight the role of institutional configurations and governance structures in shaping resilience trajectories (Adger et al., 2005; Anderies et al., 2004).

While these models offer a more realistic foundation for understanding resilience in dynamic environments, they remain difficult to institutionalize. Their complexity often leads to partial or symbolic uptake, where tools are adopted but underlying decision logics are not embedded in organizational routines or governance structures. In many cases, these difficulties are compounded by the trade-offs inherent in resilience planning, where choices about priorities and beneficiaries are shaped by political and social contexts, of-

ten constraining the feasibility and scope of resilience measures (Leichenko & O'Brien, 2024).

A.4 Emergence, Complexity, and Transformation

Resilience thinking reaches its most sophisticated form when it embraces transformation and self-organization. Complex adaptive systems, including ecosystems, urban regions, and socio-technical infrastructures, often change not just in response to shocks but because of them. These systems adapt by reconfiguring their internal structures and processes, sometimes in ways that are irreversible. Biological and ecological systems offer key examples. In biology, resilience encompasses an organism's or ecosystem's capacity to withstand and recover from environmental stressors. Evolutionary biology considers resilience through phenotypic plasticity, allowing species to adapt to changing environments, while conservation biology seeks to bolster species' resilience to threats like climate change. Physiological studies explore mechanisms that support homeostasis under stress. Coral reefs facing warming waters may shift their symbiotic algae, changing their structure to maintain function (Feder et al., 2019). In ecology, resilience shapes how systems undergo cycles of growth, conservation, release, and renewal (Gunderson & Holling, 2002). In population dynamics, these principles are modeled by the Lotka–Volterra predator–prey equations:

$$\frac{dx}{dt} = \alpha x - \beta xy \tag{1}$$

$$\frac{dy}{dt} = \delta xy - \gamma y \tag{2}$$

where x and y are prey and predator populations, respectively (Wangersky, 1978). Evolutionary game theory explores resilience through evolving strategies under shifting conditions. A strategy s^* is evolutionarily stable if $u(s^*, s^*) > u(s', s^*)$ for all $s' \neq s^*$, indicating resistance to invasion by alternatives (Gunderson & Holling, 2002; Ostrom, 1990). These logics emphasize resilience not as resistance, but as the capacity to evolve in response to disruption.

Agent-based modeling (ABM) enables the simulation of how local interactions among agents give rise to large-scale patterns of resilience. Each agent's state $s_i(t + 1) = f(s_i(t), s_j(t), E)$ evolves based on neighboring agents and environmental conditions (Helbing, 2012). ABMs are increasingly applied to organizational contexts, including simulations of decentralized coordination in disaster response, cybersecurity incidents, and interdependent infrastructures. A system is considered resilient if $\lim_{t \rightarrow \infty} s_i(t) \rightarrow S$ for all agents.

These approaches demonstrate that resilience emerges from systems composed of interacting, adaptive elements. Feedback mechanisms, redundancy, and cooperation allow complex adaptive systems to reorganize and sustain functionality after disruptions (Gao et al., 2016; Krakovská et al., 2023). In thermodynamics and statistical mechanics, resilience is associated with conditions for stability $\frac{\delta^2 \Phi}{\delta x^2} \geq 0$ and dynamic recovery: $\langle x(t) \rangle \sim \langle x_{\text{eq}} \rangle + \Delta x_0 e^{-\gamma t}$ where γ reflects the speed of stabilization (Marconi et al., 2008; Pelorosso et al., 2017).

These models highlight that resilience cannot be engineered solely through top-down design—it must be cultivated through distributed adaptation, making it inherently difficult to capture using traditional control-based frameworks. This highlights a persistent disconnect: while resilience is invoked rhetorically, few systems are designed to self-organize or evolve under pressure.

A.5 Psychological and Neurobiological Dimensions

Beyond ecological and technological systems, resilience has deep roots in the life sciences, particularly in psychology and neuroscience. In psychology, resilience describes an individual's capacity to adapt positively in the face of adversity. Research beginning with wartime family studies (Hill, 1949) and later expanded by Antonovsky on trauma and meaning (Antonovsky et al., 1971) laid the foundation for modern understandings of psychological resilience. Studies of at-risk children (Werner, 1995) demonstrated that personal traits and supportive environments jointly foster positive adaptation. Resilience is now seen as a dynamic, developmental process shaped by interactions between individual traits and external protective factors (Rutter, 1987; Ungar, 2021). Supportive social structures, belief systems, and relational bonds contribute to adaptive responses across the lifespan (Masten & Osofsky, 2010; Walsh, 1996). This capacity is not fixed but can evolve through experience and support. Recent multisystemic perspectives expand this view by emphasizing that resilience is not only psychological but also shaped by the interaction of social, institutional, and environmental systems (Ungar, 2021). In neuroscience, resilience correlates with efficient stress regulation, particularly involving the prefrontal cortex, amygdala, and hippocampus (Kalisch et al., 2024). The hypothalamic-pituitary-adrenal (HPA) axis modulates cortisol responses, and resilient individuals exhibit more

rapid recovery from stress (McEwen, 2004). Epigenetic studies suggest that life experiences can influence the expression of resilience-related genes (Rakesh et al., 2019). Cognitive strategies such as reappraisal affect neural circuitry involved in emotional regulation, linking psychological and neurobiological perspectives. Advances in computational modeling and network analysis are helping to map these pathways and predict resilience outcomes (Manikis et al., 2019). Standardized tools have helped operationalize psychological resilience. The Wagnild and Young Resilience Scale emphasizes traits like perseverance and self-reliance, while the refined 10-item Connor–Davidson Resilience Scale focuses on adaptability and emotional regulation (Campbell-Sills & Stein, 2007; Wagnild & Young, 1993). Digital technologies, including mobile assessments and machine learning, now complement traditional tools by enabling personalized monitoring and intervention (Kleitman et al., 2022).

Despite its centrality to adaptive behavior under stress, psychological resilience is rarely integrated into organizational resilience frameworks – particularly in domains like cybersecurity, where human cognition and affective regulation are critical to situational judgment. As a result, even when individual resilience is assessed or encouraged, it is often decoupled from systemic support structures, reinforcing a symbolic approach to resilience rather than embedding it in collective practice. Organizational resilience requires not only resilient individuals, but the structural and cultural scaffolding that supports cognitive and emotional adaptation under pressure.

This human-centered view of resilience emphasizes the interplay between cognitive, emotional, social, and physiological dimensions. It adds an important layer to the un-

derstanding of resilience as not only systemic or ecological, but also deeply embodied and experiential.

B SYSTEMATIC LITERATURE REVIEW PROCESS

This review followed established guidelines for structured literature reviews in Information Systems and management research (vom Brocke et al., 2009, 2015; Webster & Watson, 2002). The objective was to identify how resilience has been theorized, measured, and operationalized across disciplines, and to classify contributions according to topic and methodological approach. To ensure legitimacy, the process was evaluated along five criteria: coverage, rigor, synthesis quality, reliability, and conceptual validation.

B.1 Search and Selection

The initial search was conducted primarily on Google Scholar, which provided broad coverage across disciplines. To reinforce rigor, we also ran confirmatory searches in Web of Science and IEEE Xplore. While these yielded relatively few unique records beyond those already identified, they confirmed that no major conceptual categories were overlooked. No temporal restrictions were imposed, and only English-language sources were retained.

We began with the general keyword *resilience* and progressively refined searches by combining it with disciplinary terms (e.g., hospital, community, complex adaptive systems) and methodological terms (e.g., graph theory, pandemic, Connor–Davidson Resilience Scale). To ensure **coverage**, backward and forward citation searches were performed on highly cited and seminal works (e.g., Holling, 1973; Boh et al., 2023).

This process yielded approximately 340 records. Following scope refinement, 200 were excluded as overrepresentative or irrelevant for not meeting the inclusion criteria (e.g., purely technical, ecological without transferability, or rhetorical use), leaving 120 full-text articles for detailed eligibility assessment (see Figure A.1). Of these, 89 were identified as **core references** and 30 as **peripheral references**. **Saturation** was observed: in the final 30 screened articles, no new conceptual categories emerged. These were largely theoretical or normative surveys, which were already well represented in the corpus.

Inclusion criteria required that a study (1) explicitly address resilience, and (2) contribute to management, organizational, or operational perspectives (conceptual or empirical). Purely technical papers (e.g., infrastructure reliability models without organizational connection) or rhetorical uses of resilience without substantive analysis were excluded. Screening decisions were first made by the first author and subsequently cross-checked with co-authors, ensuring consistency and enhancing **reliability**.

B.2 Classification

Each retained study was coded along two dimensions:

1. **Topic domain:** Complex Adaptive Systems (CAS) – Engineering – Ecology & Social Systems – Climate Change & Disaster Management – Psychology & Neuroscience – Public Health – Organizational Management – Risk Management & Cyber.
2. **Research approach:** Simulation & Modeling – Network Analysis – Markers & Metrics – Technology & Behavioral Assessment – Case Studies – Longitudinal Studies – Surveys.

Papers could be coded into multiple categories when they combined disciplinary perspectives or methods, reflecting the multidimensionality of resilience. All coding decisions were documented in a shared database, creating a transparent audit trail. To further enhance **reliability**, a random subset of 20 articles was independently cross-checked by a co-author. Initial agreement was approximately 50%, reflecting the interpretive challenges of coding across disciplinary traditions. All discrepancies were subsequently discussed and resolved through iterative consensus-building among the authors. Discrepancies were documented and resolved through discussion, which ensured that final classifications were consistent and transparent.

B.3 Gap Identification

Figure A.1 presents a heatmap of the distribution of the 340 reviewed publications across topic–approach categories. The visualization demonstrates an asymmetry: simulation, network, and metrics-based approaches dominate in technical disciplines (e.g., engineering, ecology), while management, organizational, and cyber resilience research rely heavily on case studies, surveys, and qualitative descriptions. This pattern provided the empirical basis for identifying gaps.

Topic–approach intersections were coded according to their degree of coverage: gray cells denote areas excluded from scope; orange cells indicate limited coverage (few publications); and red cells mark empty categories (no identified studies). These criteria were defined during the review process and applied consistently. Importantly, the subsequent Table 2 in the main text emerged directly from these coded intersections, demonstrating

conceptual validation rather than post-hoc interpretation.

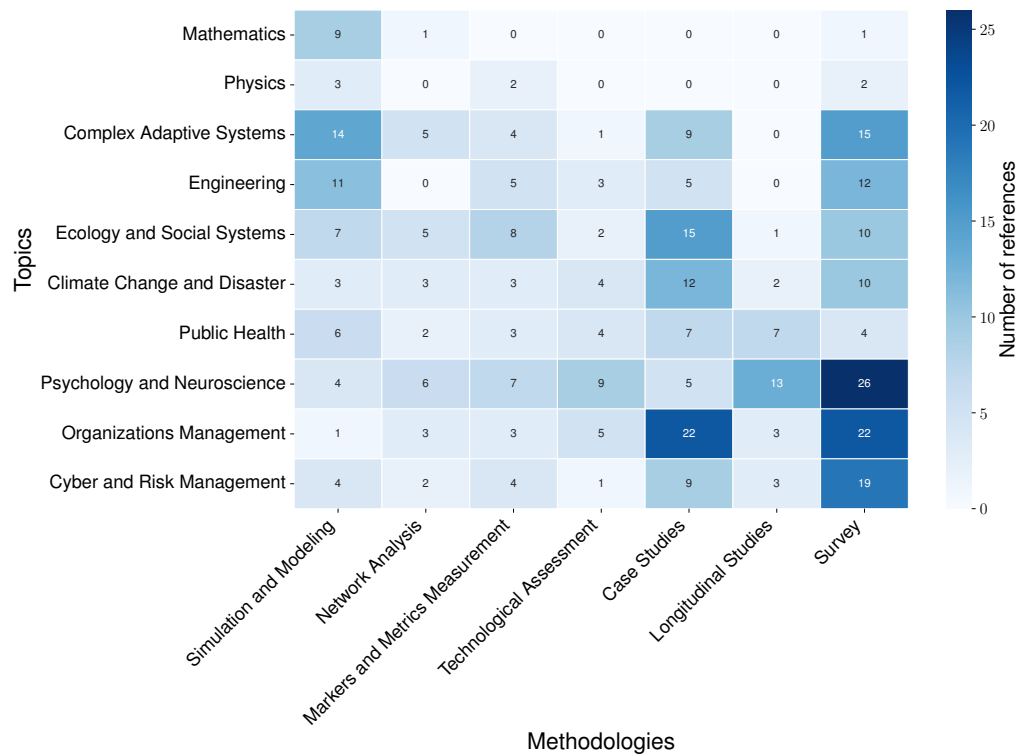


Figure A.1: Systematic Review Screening and Corpus Construction. This figure documents the staged process through which the review corpus was constructed. Across Google Scholar, Web of Science, and IEEE Xplore, 340 records were initially identified. After excluding 200 records that were overrepresentative or outside the scope of the review, 120 full-text articles were assessed for eligibility. The final analytical corpus consisted of 89 core references and 30 peripheral studies (119 total). The figure documents the screening stages and inclusion decisions, enhancing the transparency and replicability of the review.

B.4 Limitations

Two limitations must be acknowledged. First, reliance on Google Scholar ensured breadth but may underweight less-cited contributions in field-specific outlets. Future studies could extend this review with structured searches in Scopus or Web of Science. Second, classification across disciplinary and methodological categories inevitably involved interpretive judgment. However, transparency was maximized through documentation, shared coding protocols, and co-author checks. It is worth noting that the inter-coder agreement for the reliability check was approximately 50%. While this figure may appear lower than in

single-discipline reviews, it reflects the interpretive challenges of coding across diverse disciplinary traditions. To mitigate this, all discrepancies were documented and resolved through discussion, which ensured that final classifications were consistent, transparent, and conceptually robust.

B.5 Evaluation of Review Quality

In line with Webster and Watson, 2002 and vom Brocke et al., 2015, the review was evaluated along four dimensions:

- **Coverage:** Broad search terms, multiple disciplinary anchors, and citation chaining ensured inclusiveness, with evidence of saturation.
- **Rigor:** Search strategy, inclusion/exclusion rules, and coding were documented to enable replication.
- **Synthesis quality:** Rather than listing articles, the review integrated findings into topic–method categories and cross-disciplinary dimensions, providing explanatory structure.
- **Contribution:** The heatmap and Research Gaps Table offer a cumulative and actionable agenda for resilience research in IS.

Together, these elements legitimate the review as systematic, replicable, and conceptually robust.

C EMPIRICAL INSIGHTS FROM FIELD STUDY

C.1 Methodology

To complement the literature review, we draw on empirical material collected in Lorena Rota's Master's thesis (2024). Rota is a co-author of this manuscript, ensuring continuity between the thesis research and the present review. The study conducted semi-structured interviews with 18 cybersecurity and risk professionals from Swiss and international organizations. Participants were selected on the basis of their direct responsibility for resilience programs and organizational risk management.

All participants provided informed consent prior to the interviews. To ensure compliance with research ethics standards, the identities of both individuals and organizations were anonymized, and all transcripts were stored securely with access restricted to the research team.

The research protocol followed six explicit steps, adapted from Rota (2024):

1. **Definition of objectives:** Align interview objectives with the research questions and theoretical framework.
2. **Selection of participants:** Apply clear inclusion and exclusion criteria to identify relevant experts.
3. **Preparation and invitations:** Develop a detailed interview protocol, obtain informed consent, and ensure secure data management and storage.
4. **Conduct of interviews:** Use validated probing techniques during the semi-structured

interviews; record with consent or transcribe directly where recording was not authorized.

5. **Data analysis:** Apply Braun and Clarke’s systematic thematic analysis (Braun et al., 2019). No qualitative software was used; manual coding was chosen to remain close to the data.
6. **Conclusions and implications:** Link findings explicitly to raw data, and discuss both study limitations and avenues for further research.

The interview guide consisted of 19 questions, covering four domains:

- (a) how organizations define and interpret resilience,
- (b) which frameworks and standards are adopted (e.g., NIST CSF, ISO/IEC 27001, COBIT),
- (c) how governance structures allocate responsibility for resilience, and
- (d) how learning from incidents and predictive capabilities are pursued in practice.

C.2 Results

Analysis revealed six recurrent themes. These themes are synthesized in Table 1 in 1. Below, we summarize the supporting evidence.

- **Quantification and Ambiguity.** Organizations struggled to define operational metrics. Resilience was widely invoked but remained fragmented across units. One

participant noted: *“We talk about resilience in every committee, but nobody can show me how we measure it.”*

- **Regulatory Pressure and Symbolic Compliance.** Regulatory audits, such as those related to DORA, NIS2, or FINMA Circulars, dominated practice. Several interviewees described resilience efforts as primarily aimed at maintaining a “license to operate.” As one put it: *“Most of our resilience work is to show auditors we are compliant — not to prepare for real incidents.”*
- **Coordination and Silos.** IT, risk, and compliance departments operated in silos with limited cross-functional collaboration. Respondents emphasized that collaboration often occurred only after incidents: *“We only speak with risk after something happens — not before.”*
- **Learning from Incidents.** Post-incident reviews were conducted but rarely institutionalized. Reports were often archived without persistent follow-up, leading to repeated mistakes. One expert observed: *“The same issues resurface again and again because lessons learned are not retained.”*
- **Human and Cultural Factors.** Expertise, training, and organizational culture played a significant role, but interpretations varied widely across units. Simulation exercises were valued by some but dismissed by others as symbolic: *“Cyber ranges are nice exercises, but unless linked to daily practice, they remain theater.”*
- **Flexibility–Precision Paradox.** Regulators provided flexible definitions of resilience, yet firms requested precise metrics to justify investments. As an executive explained: *“We are told to be resilient, but no one explains how to measure or prove it.”*

Two concrete practices illustrate these dynamics. First, most organizations adopt established frameworks such as NIST CSF, ISO/IEC 27001, or COBIT, but they adapt them selectively to sectoral requirements rather than applying them as-is. Second, some organizations experiment with using Security Operations Center (SOC) tickets as proxies for resilience indicators, though such practices remain emergent and not standardized. Governance choices also reflect cost and efficiency concerns: rather than creating new resilience committees, most organizations integrate resilience into existing structures, which can dilute accountability.

C.3 Summary

These empirical insights demonstrate how organizations invoke resilience under regulatory and managerial pressure while struggling to embed it as an adaptive capability. They reveal six paradoxes that structure the resilience implementation gap. Table 1 in the Section 1 synthesizes these findings by aligning the themes with their organizational dimensions and implications for practice.